

We have tested the app interface by handing out the app to a small group of test users and instructing them as follows: *“Suppose you own an EV. With the provided app, book a charging session for your car.”*

Test users generally reported that personalized recommendations would be a desirable addition: if the app collects data about user behaviour and charging preferences, it can learn the kind of places users prefer to have nearby while charging, ie. cafés, restaurants, shopping areas, or grocery stores. If a user often chooses a charger close to a grocery store, the app could show similar charging stations higher in the results. This would make it possible to combine charging with everyday activities, such as doing grocery shopping while waiting for the car to charge.

The app could therefore recommend stations based on several things at the same time, such as distance, charging speed, predicted availability and nearby places. Instead of only showing the closest charger, the goal is to help the user find the charger that best fits their current situation and preferences.

In the interface, these recommendations could be shown directly in the station list or as a filter in the map and this would make the system’s learning visible to the user and help them to understand why a particular station is being recommended.

Additionally, the app can collect data such as which charging stations users normally choose, what time they charge, how long they usually charge for, and how busy the stations are at different times. This data can then be used to identify patterns and estimate how crowded a charging station might be and give the user a better idea of where they are most likely to find the charger that best fits their current situation and preferences.

Over time, the system can learn more from how people use the app. The more charging sessions and interactions the system has data from, the better the predictions and recommendations can become. In this way, the user’s interaction with the app provides feedback that can help refine future predictions and recommendations.

The main goal of this iteration is to make charging easier to plan, reduce time spent searching or waiting for a charger, and make the charging time more useful by helping users find places where they can also get coffee, eat, shop or buy groceries.

This matches the assignment’s focus on extending the system, since it uses personal usage data, users should be informed about what data is collected and how it is used, and they should be able to control whether personalized recommendations are enabled.

